Matrices as data

A grayscale image can be represented by $A \in \mathbb{R}^{m \times n}$, with pixel intensities as entries. Color images usually form tensors; a matrix treatment must specify how the channels are arranged.

Rank-one matrix:

$$uv^T = \begin{bmatrix} u_1 \\ \vdots \\ u_m \end{bmatrix} \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix},$$

$$(uv^T)_{ij} = u_i v_j.$$

A rank-one factorization stores $m + n$ scalars rather than mn entries. SVD expresses a general matrix as a weighted sum of such pieces. Large singular values often capture substantial reconstruction energy.

Statement of the SVD

Every

$$A \in \mathbb{R}^{m \times n}, \qquad \operatorname{rank} A = r,$$

admits a full SVD

$$A = U \Sigma V^T.$$

$$U \in \mathbb{R}^{m \times m}, \qquad V \in \mathbb{R}^{n \times n},$$

where U and V have orthonormal columns. The nonzero diagonal entries of the $m \times n$ matrix Σ are

$$\sigma_1 \geq \cdots \geq \sigma_r > 0.$$

Reduced SVD:

$$A = U_r \Sigma_r V_r^T,$$

$$U_r \in \mathbb{R}^{m \times r}, \quad \Sigma_r \in \mathbb{R}^{r \times r}, \quad V_r \in \mathbb{R}^{n \times r}.$$

The remaining entries of the full rectangular Σ are zero. The reduced SVD retains only the r positive singular values and their vectors.

Right singular vectors from A^TA

$$x^TA^TAx = \|Ax\|^2 \geq 0.$$

Thus A^TA is symmetric positive semidefinite and has spectral decomposition

$$A^TA = V\Lambda V^T.$$

Order the positive eigenvalues and set

$$\lambda_1 \geq \cdots \geq \lambda_r > 0,$$

$$\sigma_i = \sqrt{\lambda_i}.$$

For $i \leq r$,

$$\|Av_i\|^2 = v_i^TA^TAv_i = \sigma_i^2,$$

so $Av_i \neq 0$.

The vectors $v_1, \ldots, v_r$ form an orthonormal row-space basis. The vectors $v_{r+1}, \ldots, v_n$ form an orthonormal nullspace basis.

Left singular vectors

Define

$$u_i = \frac{Av_i}{\sigma_i}.$$

For $i \neq j$, orthogonality follows from

$$\sigma_i\sigma_ju_i^Tu_j = v_i^TA^TAv_j = \lambda_jv_i^Tv_j = 0.$$

The lengths satisfy

$$\|u_i\|^2 = \frac{\|Av_i\|^2}{\sigma_i^2} = 1.$$

Consequently,

$$Av_i = \sigma_iu_i.$$

The vectors $u_1, \ldots, u_r$ form an orthonormal column-space basis. Complete them with $u_{r+1}, \ldots, u_m$, an orthonormal basis of the left nullspace.

To complete the existence proof, let $V = [v_1, \ldots, v_n]$ be an orthonormal eigenbasis of A^TA. For $i > r$,

Left singular vectors (continued)

$$\|Av_i\|^2 = v_i^T A^T A v_i = 0$$

so $Av_i = 0$. Extend $u_1, \ldots, u_r$ to a complete orthonormal basis of $\mathbb{R}^m$. For any $x = \sum_{j=1}^{n} c_j v_j$,

$$Ax = \sum_{j=1}^{r} c_j A v_j = \sum_{j=1}^{r} c_j \sigma_j u_j = U \Sigma V^T x.$$

This verifies the right vectors, left vectors, singular values, and factorization together.

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Comparing $A^T A$ and AA^T

$$AA^T = U \Sigma \Sigma^T U^T.$$

The two Gram matrices have the same positive eigenvalues and rank. The multiplicity of zero can differ because their sizes can differ.

This explains the general structure.

- The eigenvectors of $A^T A$ give V.
- The eigenvectors of AA^T give U.
- The positive eigenvalues are the squared singular values.
- Rectangular and rank-deficient matrices also admit SVD.

Numerical implementations usually avoid forming $A^T A$, which squares the relevant condition number and can overflow. Stable methods use bidiagonal reduction and algorithms for the resulting bidiagonal singular-value problem.

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Rank-one expansion

Reduced SVD:

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T.$$

The Frobenius inner product is $\langle B, C \rangle_F = \operatorname{tr}(B^T C) = \sum_{ij} B_{ij} C_{ij}$, and $\|B\|_F = \sqrt{\sum_{ij} B_{ij}^2}$. Distinct rank-one terms are orthogonal in this inner product.

$$\langle u_i v_i^T, u_j v_j^T \rangle_F = 0, \quad i \neq j.$$

Consequently,

$$\|A\|_F^2 = \sum_i \sigma_i^2.$$

Retaining the first k terms preserves the energy fraction

$$\frac{\sum_{i=1}^{k} \sigma_i^2}{\sum_{i=1}^{r} \sigma_i^2}.$$

Rank-one expansion (continued)

This ratio requires $A \neq 0$. The nuclear norm is $\|A\|_* = \sum_i \sigma_i$, which differs from the Frobenius norm. Reconstruction energy need not match perceptual relevance, so an image application must specify its error metric and task.

Truncated SVD and Eckart–Young

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T.$$

Remainder：

$$A - A_k = \sum_{i=k+1}^{r} \sigma_i u_i v_i^T.$$

For every matrix B of rank at most k,

$$\|A - B\|_2 \geq \sigma_{k+1},$$

$$\|A - B\|_F \geq \sqrt{\sum_{i=k+1}^{r} \sigma_i^2}.$$

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Truncated SVD and Eckart–Young (continued)

For the spectral bound, B has a nonzero kernel vector in $\operatorname{span}\{v_1, \ldots, v_{k+1}\}$. Normalize it to $\|z\| = 1$. Then $\|(A - B)z\| = \|Az\| \geq \sigma_{k+1}$. For the Frobenius bound, let P project onto $C(B)$. Then $\|A - B\|_F^2 \geq \|(I - P)A\|_F^2 = \|A\|_F^2 - \operatorname{tr}(PAA^T)$. Extend the left singular vectors to an orthonormal basis $u_1, \ldots, u_m$, setting $\sigma_i = 0$ for $i > r$. Let $\alpha_i = u_i^T P u_i$. Since $0 \leq \alpha_i \leq 1$ and $\sum_{i=1}^{m} \alpha_i = \operatorname{tr} P = \operatorname{rank} P \leq k$, we have $\operatorname{tr}(PAA^T) = \sum_{i=1}^{m} \sigma_i^2 \alpha_i \leq \sum_{i=1}^{k} \sigma_i^2$. The discarded terms of A_k attain both lower bounds. Here $0 \leq k < r$; for $k \geq r$, the optimal error is zero.

If $\sigma_k = \sigma_{k+1}$, the optimal subspace need not be unique. Individual axes may rotate within a tied singular subspace.

For singular values $9, 4, 1, 0$, the rank is 3. At $k = 1$, the optimal errors are

$$\|A - A_1\|_2 = 4, \qquad \|A - A_1\|_F = \sqrt{4^2 + 1^2} = \sqrt{17}.$$

At $k = 2$, both errors equal 1; at $k = 3$, both vanish. The spectral error is the largest discarded singular value, while the Frobenius error is the square root of their squared sum.

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Storage cost of low-rank compression

Rank-k factors store

$$mk + nk + k = k(m + n + 1)$$

raw scalars, plus finite-precision metadata. Raw storage can decrease only if this count is below mn.

For a 1000×1000 image and rank 10, the factor count is 2.001×10^4, compared with 10^6 entries. Actual formats also store headers, quantization information, and other metadata.

Random full-rank data often compress poorly. Compressibility is a property of the data, not a guarantee supplied by SVD.

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The data orientation in PCA

Let

$$A_0 \in \mathbb{R}^{m \times n},$$

with variables in rows and samples in columns. With row means μ_j, center the data by

$$A_{js} = (A_0)_{js} - \mu_j.$$

The first principal direction u_1 maximizes the score vector's

$$s_1 = u_1^T A$$

sum of squares,

$$\|u_1^T A\|^2 = u_1^T A A^T u_1.$$

subject to $\|u_1\| = 1$. The spectral theorem gives maximum σ_1^2, attained at a dominant eigenvector of AA^T, the first left singular vector.

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Scores and sign ambiguity

From

$$A = U\Sigma V^T$$

the principal scores are

$$S = \Sigma V^T = U^T A.$$

Row i lists all samples' scores on component i. Its squared norm equals σ_i^2, with zero rows included in a full representation.

Changing the signs of both u_i and v_i preserves the factorization. Changing only one changes the reconstruction. An individual sign has no intrinsic meaning.

Repeated singular values permit rotations of the component axes within the tied subspace. Compare subspace angles across PCA fits rather than matching arbitrary column numbers.

Covariance denominators

For variables in rows and $n \geq 2$ samples, the usual sample covariance is

$$\widehat{C}_{n-1} = \frac{1}{n-1} AA^T,$$

The descriptive or population-denominator version is

$$\widehat{C}_n = \frac{1}{n} AA^T.$$

The eigenvectors coincide, while eigenvalues differ by factor $n/(n-1)$. For $n = 1$, the sample-variance denominator vanishes.

Unscaled PCA favors features with larger variance or larger numerical units. Standardization yields correlation-based PCA but changes the original geometry; it is a modeling choice.

A two-dimensional PCA example

Samples:

$$(-1,-1), \quad (0,1), \quad (1,0).$$

Centered matrix:

$$A = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}.$$

$$AA^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

$$u_1 = (1,1)^T/\sqrt{2}, \qquad u_2 = (1,-1)^T/\sqrt{2}.$$

Scores:

$$s_1 = \begin{bmatrix} -\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix},$$
$$s_2 = \begin{bmatrix} 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$$

The score rows are orthogonal and their squared norms are 3 and 1.

How singular vectors determine ellipse axes

The input and output each use an orthonormal basis, generally with different directions. Take

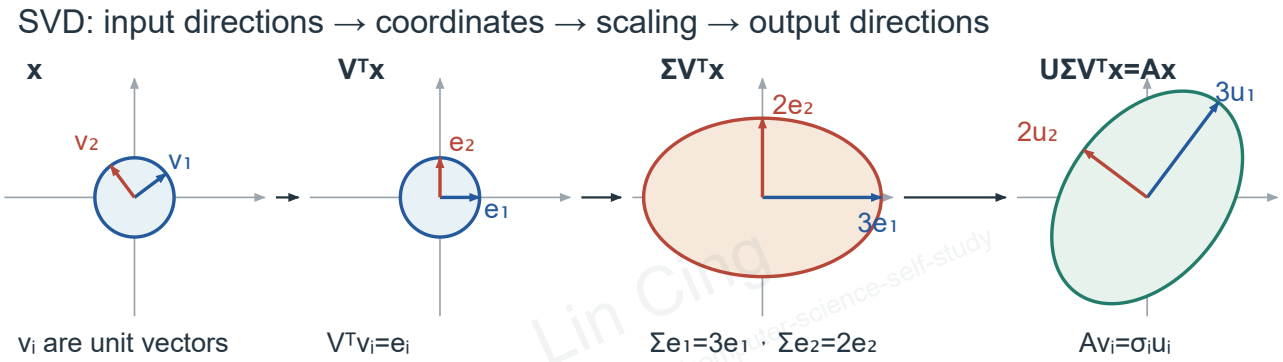
$$V = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}, \qquad U = \frac{1}{5}\begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}, \qquad \Sigma = \operatorname{diag}(3,2).$$

Here $V^TV = U^TU = I$, and

$$A = U\Sigma V^T = \frac{1}{5}\begin{bmatrix} 12 & -1 \\ 6 & 12 \end{bmatrix}.$$

All panels use the same coordinate scale. First V^T maps input directions v_i to e_i, then Σ scales, and finally U sets the output directions. Rotation preserves the circle as a set, but moves the marked vectors; that step must remain visible.

How singular vectors determine ellipse axes (continued)



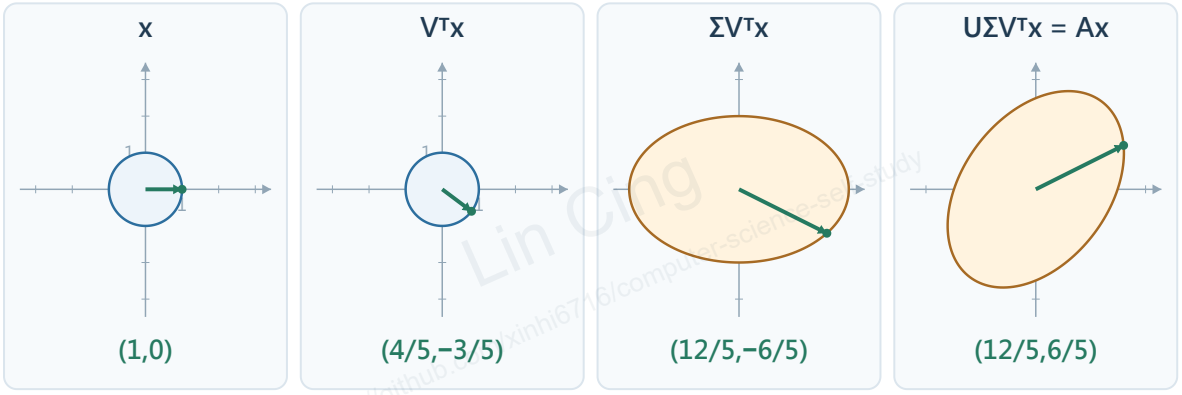
Check the two axes individually.

$$v_1 = (4/5, 3/5)^T, \quad Av_1 = (9/5, 12/5)^T = 3u_1,$$
$$v_2 = (-3/5, 4/5)^T, \quad Av_2 = (-8/5, 6/5)^T = 2u_2.$$

The semiaxis vectors are $3u_1, 2u_2$. The unit vectors u_1, u_2 specify directions. This full-rank example maps the unit circle to an ellipse boundary and the unit disk to the filled elliptical disk.

Track a unit vector outside the singular directions

Keep the preceding U, V and Σ, and take $x = (1, 0)^T$, which is neither singular direction. The green arrow tracks this point; each background curve is the image of the unit circle at that stage.



Same scale in every panel; the green vector is the same input tracked through the map.

Check coordinates and lengths separately

Stage	Resulting coordinates	Squared norm
Input x	(1,0)	1
$V^T x$	(4/5,−3/5)	1
$\Sigma V^T x$	(12/5,−6/5)	36/5
$U\Sigma V^T x = Ax$	(12/5,6/5)	36/5

V^T and U preserve lengths; Σ performs the scaling. Here $\|Ax\| = 6/\sqrt{5}$, which is neither singular value.

For any unit input, write $w = V^T x$. Then $w_1^2 + w_2^2 = 1$, and

$$\|Ax\|^2 = 9w_1^2 + 4w_2^2 = 4 + 5w_1^2 \in [4, 9].$$

Thus $2 \le \|Ax\| \le 3$, with the upper bound at $x = \pm v_1$ and lower bound at $x = \pm v_2$. This argument covers every unit input; one numerical trace alone would not prove the bounds.

SVD geometry with a right-angle rotation

For the unit ball $\|x\| \le 1$,

$$y = Ax = U\Sigma V^T x.$$

all three steps use the same v_i, u_i. Take

$$V = I, \quad \Sigma = \mathrm{diag}(3, 2), \quad U = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad A = U\Sigma V^T = \begin{bmatrix} 0 & -2 \\ 3 & 0 \end{bmatrix}.$$

Apply the factors in order.

1. $V^T x = Ix$ leaves input coordinates unchanged; the disk retains the coordinate-axis directions.
2. $\Sigma(V^T x) = (3x_1, 2x_2)$ scales the orthogonal axes, producing semiaxes 3 and 2.
3. $U\Sigma V^T x$ rotates the ellipse, sending the standard axes to $u_1 = (0, 1)^T, u_2 = (-1, 0)^T$.

The definitions give

$$v_1 = (1, 0)^T, \quad Av_1 = (0, 3)^T = 3u_1,$$
$$v_2 = (0, 1)^T, \quad Av_2 = (-2, 0)^T = 2u_2.$$

SVD geometry with a right-angle rotation (continued)

Thus the ball maps to a filled ellipsoid in $C(A)$, with positive singular values as semiaxis lengths. In full-rank two dimensions, its boundary is an ellipse. With a nullspace, the unit sphere can map to the filled ellipsoid rather than only its surface. The maximum singular value is

$$\|Av_1\| = \sigma_1.$$

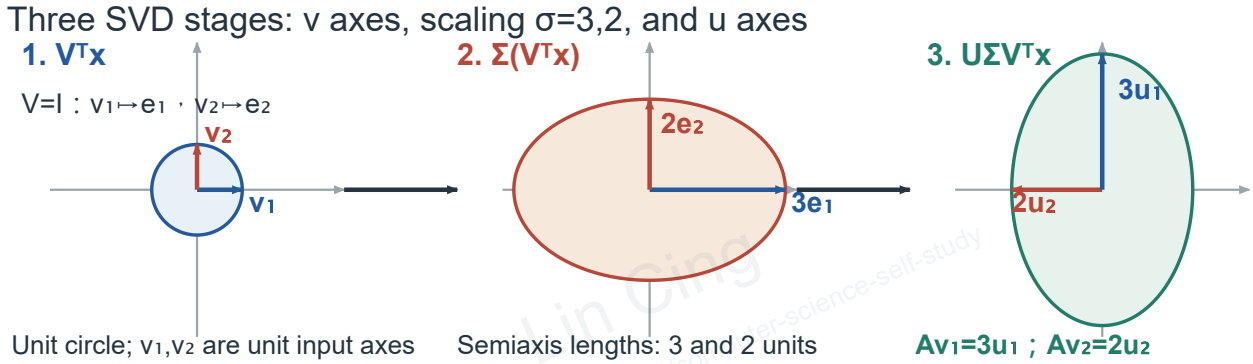
Matrix norm：

$$\|A\|_2 = \sigma_1.$$

The smallest positive σ_r controls the weakest growth from row space to column space.

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SVD geometry with a right-angle rotation (continued)



For deficient rank, take

$$B = \begin{bmatrix} 0 & 0 \\ 3 & 0 \end{bmatrix} = U\,\mathrm{diag}(3,0)V^T.$$

Then $Bv_1 = (0,3)^T$, while $Bv_2 = 0$. The v_2 direction collapses, and the unit disk maps onto the complete segment from $(0,-3)^T$ to $(0,3)^T$.

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SVD geometry with a right-angle rotation (continued)

Rank-deficient SVD: B=[[0,0],[3,0]], $\sigma_2=0$ collapses



rank(B)=1; the full image is a line, the circle image a segment

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Spectral norm and condition number

$$\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}.$$

Expand

$$x = \sum_i c_i v_i,$$

$$\|Ax\|^2 = \sum_i \sigma_i^2 |c_i|^2 \leq \sigma_1^2 \|x\|^2.$$

Equality holds at $x = v_1$, so

$$\|A\|_2 = \sigma_1.$$

For full column rank, the usual singular-value condition number is

$$\kappa_2(A) = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

An orthogonal square matrix has every singular value equal to 1 and condition number 1. A rank-deficient matrix has a zero smallest singular value and no finite ordinary inverse condition number.

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Distance to singularity and numerical rank

For an invertible square $A \in \mathbb{R}^{n \times n}$, remove $\sigma_n u_n v_n^T$.

$$\|A - A_0\|_2 = \sigma_n.$$

Thus σ_n is the spectral-norm distance to singularity. More generally, A_k is a nearest rank-at-most-k matrix, at distance σ_{k+1}.

Numerical rank uses a tolerance τ.

$$\sigma_i > \tau \|A\|.$$

The tolerance must reflect noise, dimensions, and downstream sensitivity. Stable singular values do not imply stable individual singular vectors, especially near repeated or closely spaced values.

Perturbation radius and numerical rank

For equally sized A, B, let $\delta = \|A - B\|_2$. Eckart–Young gives $\sigma_i(A) = \min_{\operatorname{rank} C < i} \|A - C\|_2$. Use the best low-rank approximation of B and the triangle inequality.

$$\sigma_i(A) \le \|A - B\|_2 + \sigma_i(B).$$

Exchanging A, B proves Weyl's bound $|\sigma_i(A) - \sigma_i(B)| \le \delta$. This is an absolute bound; relative errors in tiny singular values may still be large.

Suppose observed A is within δ of an unknown true B, and let $r_\delta = \#\{i : \sigma_i(A) > \delta\}$. Every compatible B has rank at least r_δ, since those singular values stay positive. Conversely, A_{r_δ} has error $\sigma_{r_\delta + 1} \le \delta$, so a compatible rank-$r_\delta$ model exists. If $r_\delta = q$, use $A_q = A$.

Thus r_δ is the minimum rank guaranteed within the perturbation ball, not the exact rank of every compatible model. For $A = \operatorname{diag}(1, 10^{-8}, 10^{-12})$, $\delta = 10^{-10}$, both a rank-2 truncation and the original rank-3 matrix are compatible. The data cannot identify whether the third direction is exactly zero.

Perturbation radius and numerical rank (continued)

For a fixed absolute threshold t, classifications are robust only when $\sigma_i(A) > t + \delta$ or $\sigma_i(A) < t - \delta$. Directions near the threshold are uncertain. A relative threshold $t = \tau \|A\|_2$ also requires a recorded scale and noise model.

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QR pivots are not singular values

In exact arithmetic, $A = Q_1 R$, $Q_1^T Q_1 = I$, gives $A^T A = R^T R$, so A and R share singular values. However, the diagonal of R is not its singular-value list. Take

$$R = \begin{bmatrix} 1 & K \\ 0 & 1 \end{bmatrix}, \qquad K \geq 1.$$

Both diagonal entries are 1, but $\|Re_2\|_2 = \sqrt{K^2 + 1}$, giving $\sigma_{\max} \geq K$. Since $\sigma_{\max}\sigma_{\min} = |\det R| = 1$, we obtain $\sigma_{\min} \leq 1/K$ and $\kappa_2(R) \geq K^2$. A large off-diagonal entry can conceal near dependence from the diagonal alone.

The sizes of R_{ii} can be a heuristic, not a singular-value perturbation theorem. Column-pivoted QR often estimates numerical rank, but guaranteed accuracy requires explicit rank-revealing bounds or an SVD check. A nonzero floating-point pivot need not represent a trustworthy data direction.

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Polar decomposition

$$A = U\Sigma V^T = (UV^T)(V\Sigma V^T).$$

This page concerns square $A \in \mathbb{R}^{n\times n}$, so U, V, Σ all have size $n \times n$. Set

$$Q = UV^T, \qquad S = V\Sigma V^T.$$

Then Q is orthogonal and S symmetric positive semidefinite, giving

$$A = QS.$$

If A is invertible, every singular value is positive, so S is positive definite. The sign of $\det Q$ records orientation reversal. The factors separate an orthogonal rotation or reflection from a symmetric stretch.

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The pseudoinverse

Reduced SVD:

$$A = U_r\Sigma_r V_r^T.$$

Moore-Penrose pseudoinverse:

$$A^+ = V_r\Sigma_r^{-1}U_r^T.$$

$$AA^+ = U_rU_r^T, \qquad A^+A = V_rV_r^T.$$

These are the orthogonal projectors onto column space and row space, respectively.

Full column rank:

$$A^+ = (A^TA)^{-1}A^T.$$

Full row rank:

$$A^+ = A^T(AA^T)^{-1}.$$

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The pseudoinverse (continued)

The vector A^+b is the minimum-norm least-squares solution. To prove this, write $x = \sum_i c_i v_i$ in the full right singular basis. Then $\|b - Ax\|^2 = \sum_{i=1}^{r}(u_i^T b - \sigma_i c_i)^2 + \sum_{i=r+1}^{m}(u_i^T b)^2$. Minimization requires $c_i = u_i^T b/\sigma_i$ for $i \le r$. Nullspace coefficients remain arbitrary. Setting them to zero uniquely minimizes $\|x\|^2 = \sum_i c_i^2$, giving A^+b.

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Rank-deficient least squares

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} = (1,2)^T(1,1).$$

Normalize:

$$u_1 = (1,2)^T/\sqrt{5}, \qquad v_1 = (1,1)^T/\sqrt{2}.$$
$$A = \sqrt{10}\, u_1 v_1^T.$$

Here u_1 is an output direction and v_1 an input direction. Both have two coordinates in this example but have different values. For a rectangular matrix even their dimensions can differ. Rank one does not imply $u_1 = v_1$.

For $b = (1,3)^T$,

$$A^T A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 7 \\ 7 \end{bmatrix}.$$

Every parameter satisfying

$$x_1 + x_2 = 7/5$$

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Rank-deficient least squares (continued)

is a minimizer. The normal matrix is singular and cannot be inverted.

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The minimum-norm solution

$$\sigma_1 u_1^T b = \sqrt{10}\frac{1+6}{\sqrt{5}} = 7\sqrt{2}.$$

$$x^+ = A^+ b = \frac{1}{\sigma_1} v_1 u_1^T b = \frac{7}{10}(1,1)^T.$$

Projection:

$$AA^+ b = u_1 u_1^T b = \frac{7}{5}(1,2)^T.$$

Residual:

$$e = b - AA^+ b = \left(-\frac{2}{5}, \frac{1}{5}\right)^T.$$

$$e^T(1,2) = 0,$$

The residual is therefore orthogonal to column space.

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Numerical workflow

SVD workflow:

1. Check shape, missing values, and units.
2. Center PCA data and standardize when the modeling choice requires it.
3. Use a stable SVD algorithm without forming normal equations.
4. Inspect singular-value gaps and numerical rank.
5. Choose k from the error budget.
6. Validate on held-out data or the downstream task.

Signs can flip and tied axes can rotate. Linear reconstruction alone establishes neither causal importance nor fairness.

The four subspaces

Space	Ambient space	SVD basis	Dimension
row space	$\mathbb{R}^n$	$v_1, \dots, v_r$	r
nullspace	$\mathbb{R}^n$	$v_{r+1}, \dots, v_n$	$n - r$
column space	$\mathbb{R}^m$	$u_1, \dots, u_r$	r
left nullspace	$\mathbb{R}^m$	$u_{r+1}, \dots, u_m$	$m - r$

Input and output singular vectors belong to different roles and cannot generally be paired by an inner product. Equal dimensions allow an explicit coordinate identification, but do not make the bases identical. SVD pairs them through

$$Av_i = \sigma_i u_i.$$

Chapter check

You can now do the following.

- Construct v_i from $A^T A$ and set $u_i = Av_i/\sigma_i$.
- Complete the null directions to obtain a full SVD.
- Compute spectral and Frobenius truncation errors.
- Explain PCA centering, scores, variance, and denominator choices.
- Distinguish input and output singular bases.
- Derive the operator norm, condition number, and distance to singularity.
- Construct polar factors and a pseudoinverse.
- Handle sign ambiguity, ties, and numerical rank.

Chapter 8 reinterprets transformations through bases and coordinates.

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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